SIMULTANEOUS GO VIA QUANTUM COLLAPSE

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Abstract. We give one definition of simultaneous games and simultanezation of a deterministic game with two players. As the main motivation and example, we construct a simultaneous game SGo, which is a simultanezation of Go. SGo is in some ways a simpler game than Go, as Komi and Ko rules are removed. On the other hand the states are more complex, because there are "entanglement phenomena", an idea inspired by quantum mechanics. However there are no dice rolls, instead there is a certain deterministic "quantum state" reduction, so that state evolution is deterministic. We show that SGo has a strategy to draw on average, but it is an open problem whether it has a strategy to draw. Another open problem is whether chess has a simultanezation.

1. Introduction

The game Go is an ancient and interesting game of local to global geometric principles. Like all sequential games it suffers from time asymmetry between the players Black and White. In Go this is addressed by an ad hoc rule called Komi: whereby the player White which moves second is given a specific point advantage. The trouble of time asymmetry goes deeper. For Go, it results in counter-intuitive local pathologies in game states, for example Ko situation, where after a capture of a stone the opposing player may immediately recapture, but is forbidden by an extra ad-hock rule, to avoid an infinite loop situation.

At the same time, precisely because of its local to global nature Go is "almost time symmetric". We make this mathematically precise by constructing a simultaneous game SGo that is in one mathematical sense a simultanezation of Go. ***SGo*** is a deterministic game, which is totally playable on a standard Go board, although we need a third type of stones that we call red. SGo is in some ways a simpler game than Go as it loses the Komi and Ko rules, and it has just two rules stone placement and capture (aside from end of game rule), but the latter and game states are now more complicated, as we have to deterministically resolve simultaneous action. The idea for SGo is inspired by quantum mechanics. Probably the most interesting question at the moment is whether SGo has a strategy to draw. We partly discuss this at the end of the paper.

Some fundamental advances in machine learning recently lead to computer Go agents winning against stellar human opponents, particular alphaGo winning against Lee Sedol. At the same time computer agents do much worse in games with imperfect information, like Poker. [1] It is interesting to speculate how alphaGo and its cousins would do against a human opponent in SGo, which is both more complex in its state space and is an imperfect information game.

[1] At least in Poker it seems that computer agents have caught up as of 2025.

2. Game rules

The game is initialized with an empty standard Go board, with virtual horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in Go, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in SGo they do this on the same turn without knowledge of the others move on this turn. This requires modifying the placement and capture rules of Go, while removing the Komi rule, and Ko rules. We also need to introduce a third type of stones called red, which may be informally interpreted as simultaneously white and black.

2.1. Some definitions.

Given a board game state S, that is any collection of white, black, red stones on the board let $G(S)$ denote the graph with vertices stones, and edges between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency, always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A colored component is a connected component of the full sub-graph of $G(S)$ whose vertices are either all the black, or all the white stones. We call this a white,respectively black component. (Full sub-graph means that the inclusion map of the sub-graph induces a bijection on sets of edges.)

As in classical Go, a connected component C has no liberties if it cannot be extended (as a connected component) by placing a stone on the board. We will also call C a trapped component. We define stones adjacent to C, as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

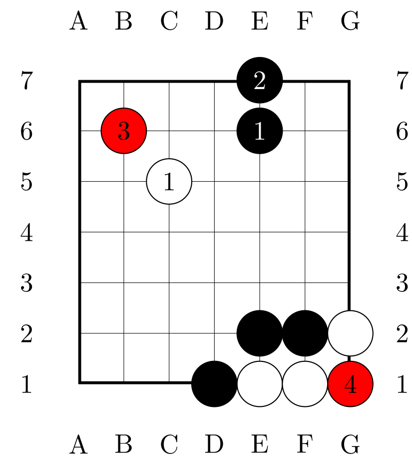
Given a colored component C, A white/black stone placed at an intersection is called a trapping if it is adjacent to a trapped colored component of opposite color.

2.2. Stone placement resolution rule of SGo.

If the players move to the same intersection on the board we place a stone there whose color resolves according to the following. If either a white or black stone placed at the intersection is trapping, then a red stone is placed at the intersection. We call this a trapping entanglement. Otherwise, if the move is not a trapping trapping entanglement we have the following:

(1) If the intersection is adjacent to a black stone, and neither a white stone nor red stone, a black stone is placed at the intersection.
(2) If the intersection is adjacent to a white stone, and neither a white stone nor red stone a white stone is placed at the intersection.
(3) Otherwise, a red stone is placed.
(4) If the stone placed by Black/White is not trapping, but is contained in a trapped colored component, it is ruled a suicide, and is an illegal move.

To the right is an example of placement resolution. Note that the stone placed on move 4 does not resolve to white, as it is trapping.



2.3. Capture rule of SGo. After both White and Black move on current turn numbered n, using the stone placement rule above, we get some board state S.

The stage 1 capture rule is then invoked on this turn as follows. Let C be a colored component with no liberties of the board state S, (the stones of C may have been placed on this turn.) Suppose that:

(1) There is no red stone adjacent to C placed on turn n.
(2) Some stone adjacent to C was placed on turn n.

Then we call C trapped on turn n, the stones of C are also called trapped on turn n turn. Suppose that no stone adjacent to C was trapped on this turn. Then C is resolved as captured.

In the case C is captured, supposing C is white:

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely are replaced by black stones.)

If C is black the process is reflected. The above stage 1 capture rule is invoked recursively until no captured colored components remain. We call this stage one reduction and the resulting board state will be denoted by $R_n(S)$.

Lemma 1. The stage one reduction is well defined, that is independent of the order in which killed components are captured.

Proof. Suppose we have two colored components C_0, C_1 captured on the given turn. By assumptions no stones in C_1 can be adjacent to stones of C_0, and vice versa. Hence applying capture rule to the group C_0 and then to group C_1, results in the same board state as first applying the capture rule to the group C_1 and then the group C_0. The result clearly follows. □

If there are no colored components trapped on the turn $n-1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction denoted by $R^k(S)$ satisfies:

$$R^1(S) = R_n(S).$$
$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

If there are no colored components trapped on the turn $n-k$ or lower, the turn is resolved after stage k reduction.

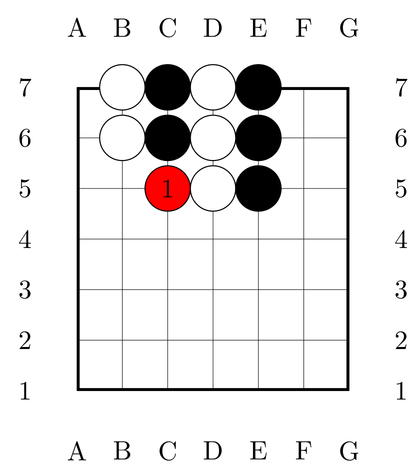
Remark 2. This may appear to be fairly complicated, but in practice the resolution of most turns is easy to work out mentally, we will give some examples of such "nested entanglement states" further ahead. Of course having a computer work out turn resolution automatically is very helpful.

2.4. End of the game. As in Go, SGo game ends after both players pass. At which point the winner is decided as in Go by territory and prisoner count. White's territory boundary consists of connected components white/red stones. Blacks territory boundary consists of connected components of black/red stones. Red stones are never counted as prisoners. A draw is now possible although compared to chess it might be much less likely, (on a typical 13 ×13 or 19 × 19 go board). See Section 3.2 for an example.

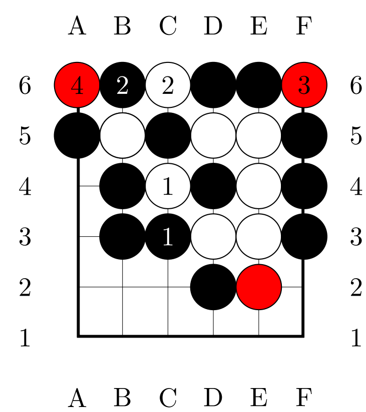
3. Examples

We give here some examples. In most of these example, the players need only play the position without needing to hedge against the moves of the opponent. In more marginal, complex positions there are such mind games, and search for a game theory optimal hedging strategy. See however Question 2.

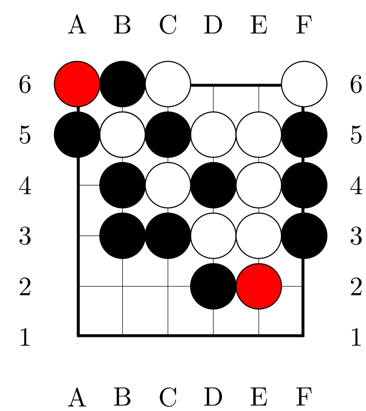
On move 1 both Black and White move to C5. The left pair of black stones will be captured next turn.



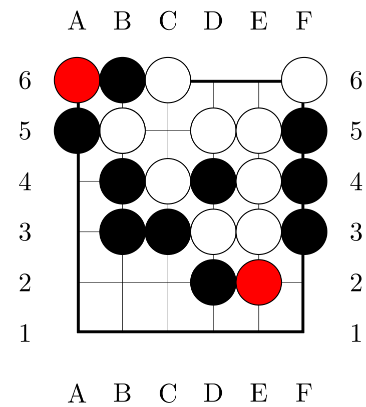
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. This gives an example of a “nested entangled state”. A possible continuation is indicated.



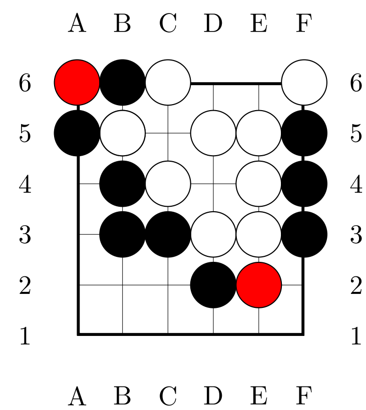
At the end of turn 4, stage one resolution is depicted on the right.



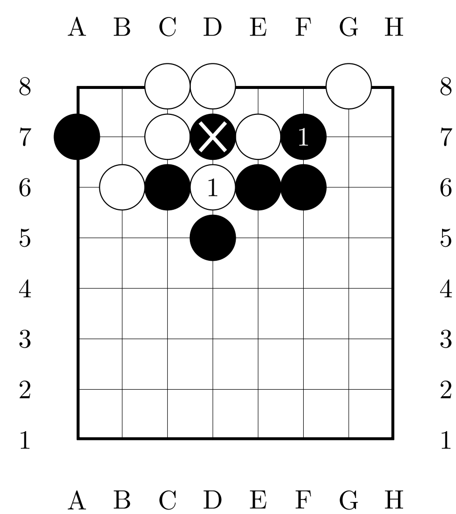
Stage two resolution is depicted on the right.



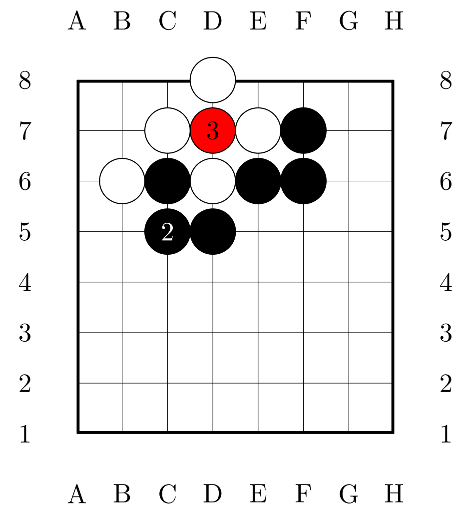
Stage three and final state resolution is depicted on the right.



Ko situation. Recall, that we don't have a Ko rule, but it is now clearly unnecessary as we can't have any infinite loops. See example on the right. White's D6 is a capture. Note that this is already implicitly in the rules set, because the white stone at D6 is not trapped on this turn. By the rules, to be trapped on this turn a stone cannot be created on this turn. This behavior is chosen for backwards compatibility with Go, and for the purpose of Theorem 1 in particular.

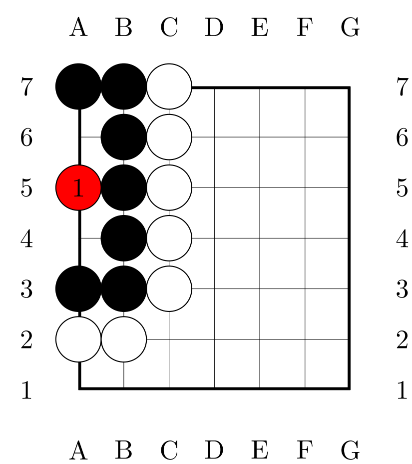


Here is a possible continuation.

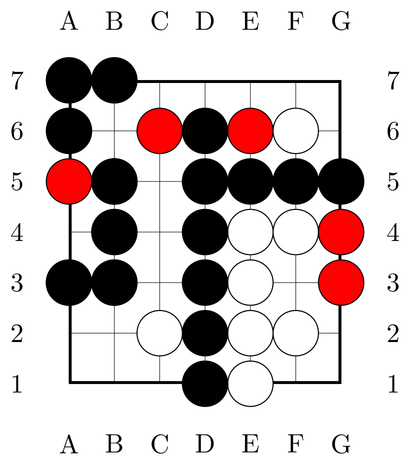


3.1. Life and death. Life and death is positionally very similar to Go, here is an example.

The Black group is alive, that is impossible to kill.



3.2. End game position. Both players pass and the game ends. Assuming there are no prisoners. White has 4 points, Black has 19 points.



4. SGo as a simultaneazation of Go

We give here a general definition of simultanezation of a deterministic game, and show that *SGo* is a simultanezation of *Go*. The result is not unsurprising, for example chess likely does not have a simultanezation in our sense, although formal verification of this is not obvious. First we give some basic definitions. We will treat (finite) deterministic games as certain types of (finite) deterministic automata.

Definition 1. A (finite) a deterministic game G with 2 players P_0, P_1 or Black, White, consists of:

- A (finite) deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- Nonempty subsets $W_0(G), W_1(G), D(G) \subset S(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S(G)$, understood as the state in which the game is initialized.
- A $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.
- A partial game evolution function:
$$E = E_G : S(G) \times M(G) \to S(G).$$

We recursively define:
$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$

From now on we restrict to finite deterministic games for simplicity.

Any game G can be naturally augmented with formal pass moves, that we call shifts. We say that G as above has shift moves if there are distinguished elements $p_0 \in M_0(G)$, $p_1 \in M_1(G)$ s.t. for all $s \in S(G)$ either $E(s, p_0)$ is defined or $E(s, p_1)$ is defined. If $E(s, p_0)$ is defined then $E(s, p_0, p_1) = s$. Similarly if $E(s, p_1)$ is defined then $E(s, p_1, p_0) = s$. Given G we denote by $\overline{G}$ its augmentation in this sense.

4.1. Simultaneous games. Simultaneous games are an extremely well studied subject, due to their prominence in economic theory, but our simultaneous games have state evolution, and so we have to set up some notions and notation to deal with this.

Definition 2. A game G as above is called simultaneous if the following conditions are satisfied:

(1) The finite set $S(G)$ has a distinguished representation as
$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$
where $B(G)$ is some set. ($B(G)$ may be interpreted as the set of "board states").

(2) $q_0 \in B(G)$ and $W_0(G), W_1(G), D_G \subset B(G)$.

(3) The function E has the properties: if $s \in B(G) \subset S(G)$, then $E(s,m) = (s,m)$ for all $m \in M(G)$ s.t. $E(s,m)$ is defined. If $s \in B(G) \times M_0(G)$, and $m \in M_1(G)$ then $E(s,m) \in B(G)$ (or undefined). If $s \in B(G) \times M_1(G)$, and $m \in M_0(G)$ then $E(s,m) \in B(G)$ (or undefined). E is undefined in all other cases.

(4) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s,m_0)$ is defined and $E(s,m_1)$ is defined then $E(s,m_0,m_1)$ is defined.

(5) For s, m_0, m_1 as in the previous property, if $E(s,m_0,m_1)$ is defined then so is $E(s,m_1,m_0)$ and
$$E(s,m_0,m_1) = E(s,m_1,m_0).$$

Note that if G is simultaneous by definition above it merely means that its state evolution rules allow it to be played with simultaneously played moves. But it may also be played with White and Black taking turns, and also knowing each other's move.

Definition 3. A game sequence in G is a finite sequence $\{m_j\}_{j=1}^{j=k}$ in $M(G)$, s.t. $E(q_0, m_1, \ldots, m_{i-1}, m_i)$ is defined for all $1 \leq i \leq k$, and
$$E(q_0, m_1, \ldots, m_k) \in W_0(G), W_1(G), D(G).$$

We then say that $\{m_j\}_{j=1}^{j=k}$ is winning for Black, respectively winning for White, respectively drawing.

For a simultaneous game G, k above must be even, and any game sequence $\mathfrak{s} = \{m_j\}_{j=1}^{j=2k}$ may be rewritten in the form $\{(m_j^0, m_j^1)\}_{j=1}^{j=k}$ where m_j^0, m_j^1 are $M_0(G)$, $M_1(G)$ respectively. A serialization of $\mathfrak{s}$ is some assignment to each pair (m_j^0, m_j^1) one of the following ordered tuples of moves in $\overline{G}$:

- m_j^0, m_j^1.
- m_j^0, p_1.
- p_0, m_j^1.
- p_0, m_j^1, m_j^0, p_1.

The juxtaposition of these tuples, naturally ordered by the order j of pair (m_j^0, m_j^1) they come from, gives a formal sequence of elements of $\overline{G}$ we denote by $\mathrm{ser}(\mathfrak{s})$, that depends on the chosen serialization. For example if $\mathfrak{s} = \{(m_1^0, m_1^0), (m_2^0, m_2^1)\}$ and the first tuple is m_1^0, m_1^1, and the second tuple is p_0, m_2^1, m_2^0, p_1 then $\mathrm{ser}(\mathfrak{s}) = \{m_1^0, m_1^1, p_0, m_2^1, m_2^0, p_1\}$.

Definition 4. Let G_0, G_1 be games, with G_1 simultaneous. We say that G_1 is a simultanezation of G_0 if the following is satisfied.

(1) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
(2) If $\mathfrak{s}$ is a game sequence in G_1, there is a serialization so that $\operatorname{ser}(\mathfrak{s})$ is is a game sequence in $\overline{G}_0$. Furthermore, if $\mathfrak{s}$ is winning for Black, White, drawing then $\operatorname{ser}(\mathfrak{s})$ is winning for Black, White, drawing respectively.
(3) There is a map $\phi : S(G_1) \to S(G_0)$ s.t. $E_{G_1}(s,m)$ is defined if and only if $E_{G_0}(\phi(s), m)$ is defined.

Theorem 1. SGo is a simultanezation of Go.

Proof. We first outline the formal description $G = SGo$ as a simultaneous game, based on the description in Section 2. The space $M(G)$ is the same as Go, we suppose that $M_0(G)$ corresponds to moves of Black. The state space is:

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where $B(G)$ is the set of all legal board positions as in the description in Section 2, meaning that the position is already resolved with respect to the movement resolution, and capture rule of SGo.

The evolution map E is then obviously defined. For example if $b \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves then $E((b, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position, using the move placement and capture rules of Section 2.3, after Black's move m_0, and White's move m_1 are reflected on the board.

Property 2 essentially follows by construction of SGo. To verify the final property, define $\phi : S(G) \to S(Go)$ by the following conditions. If $s \in B(G)$ then $\phi(s)$ is the Go board state obtained by replacing every stone on the board by a black stone. If $s \in B(G) \times M_i(G)$ then $\phi(s) = \phi(b)$, where $b = pr(s)$, for $pr : B(G) \times M_i(G) \to B(G)$ the projection. It is then immediate that Property 3 is satisfied. □

Question 1. Does chess have a simultanezation?

It is not too hard to see that any deterministic game G has a "formal simultanezation" satisfying the first two properties in the definition above. The construction of this is somewhat involved and will be omitted. However, property 3 will not be satisfied in the case of chess, and it is hard to imagine that it could simultaneously be satisfied, the issue being non-locality of chess state evolution.

4.2. Solvability. A theorem of Zermelo [2] says roughly that in every sequential game, either Black or White has a winning strategy or there is a draw strategy. Obviously SGo cannot have a winning strategy.

Question 2. Does SGo have a strategy to draw?

We point out that there are trivial examples of simultaneous games with a strategy to draw. For a very trivial example, take any finite simultaneous game G and reset $D(G) = B(G) - q_0$, $W_0(G) = \emptyset$, $W_1(G) = \emptyset$. Of course we have simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. But the latter has a strategy to draw on average. The latter being a very trivial case of the existence of the Von-Neumann, Nash equilibrium for "matrix payoff games", [1]. SGo is not a matrix payoff game. However, Nash's proof of the equilibrium theorem readily extends to this case. In general:

Theorem 2 (Nash). Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.

Proof. For $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \,|\, E(s,m) \text{ is defined}\}.$$

The space of strategies of player P_i is then:

$$\sqcup_{s \in B(G)} \mathcal{D}(M_i(s)),$$

where $\mathcal{D}(M_i(s))$ is the space of probability distributions over the finite set $M_i(s)$. Since $B(G)$ is also finite the strategy space is compact.

The argument of Nash [1] using Kakutani's fixed point theorem then readily extends to get existence of an equilibrium point. □

Corollary 1. SGo has a strategy to draw on average.

Proof. Set $G = SGo$. Let $R : S(G) \to S(G)$ be the map that interchanges the black and white colors of the stones of the corresponding board states. And let $R_M : M(G) \to M(G)$ likewise interchange black and white moves. Clearly, $E(R(s), R_M(m)) = R(E(s,m))$. In other, words state evolution is symmetric for Black and White.

Assign 1 for a win, -1 for a loss and 0 to a draw. Then it is enough to show that for any Nash equilibrium strategy the expected payoff is 0 for both players. Suppose otherwise, and suppose without loss of generality that $\mathcal{S}$ is an equilibrium strategy where Black has a positive expected payoff. Applying the reflection operators R, R_M to the strategy $\mathcal{S}$ we get that there is an equilibrium strategy where White has positive expected payoff. But this is absurd. □

References

[1] J. F. j. Nash, Equilibrium points in n-person games, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49.

[2] U. Schwalbe and P. Walker, Zermelo and the early history of game theory, http://www.math.harvard.edu/~elkies/FS23j.03/zermelo.pdf.